

Operations Research

Lecture 5: Simplex Tableau and Sensitivity Analysis

A textbook treatment

Based on the lecture notes of El Mehdi Saad

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How to read this book

Each concept in this book is presented in the same rhythm.

- 1. Motivation and intuition.** Why do we care? What problem does the idea solve, and what picture should you hold in your head?
- 2. Formal statement.** The precise definition or theorem, set in a coloured box so it is easy to find when revising.
- 3. Proof.** Every claim is proved. Longer proofs are split into labelled steps, and the one idea that makes the proof work is isolated in a *Key insight* box.
- 4. Interpretation and discussion.** What the result really says, where it applies, and which misreadings to avoid.
- 5. Examples.** From “trivially small” to “genuinely useful.”
- 6. Techniques, classic problems and gotchas.** Computational shortcuts, proof patterns, exam favourites with one-line solutions, and the mistakes everyone makes once.

Margin notes carry the informal, “say it out loud” version of what is happening in the main text. Every major section closes with a summary box.

Prerequisites. Comfort with vectors, matrices, systems of linear equations, row reduction and the notions of rank and linear independence. Chapter 2 recalls exactly what is needed as it goes.

Chapter 5

The Simplex Tableau in Matrix Form and Sensitivity Analysis

5.1 A Simplex Refresher, and the Matrix Behind It

Motivation

Chapter 3 ran the simplex method on tableaux, and Chapter 4 showed that the final objective row hides a dual solution. This chapter opens the tableau up. Part one rewrites a simplex move in matrix language: a *basic direction*, its *step length* and its *gain*, the reduced cost. This yields an optimality theorem with a two-line proof. Part two uses the same matrices for *sensitivity analysis*: when a datum of an already-solved LP changes, does the old optimal basis survive, what is the new value, and how do we continue from the last tableau instead of starting over?

The tableau is not just bookkeeping. Every number in it is a piece of B^{-1} , and the whole chapter is about reading those numbers.

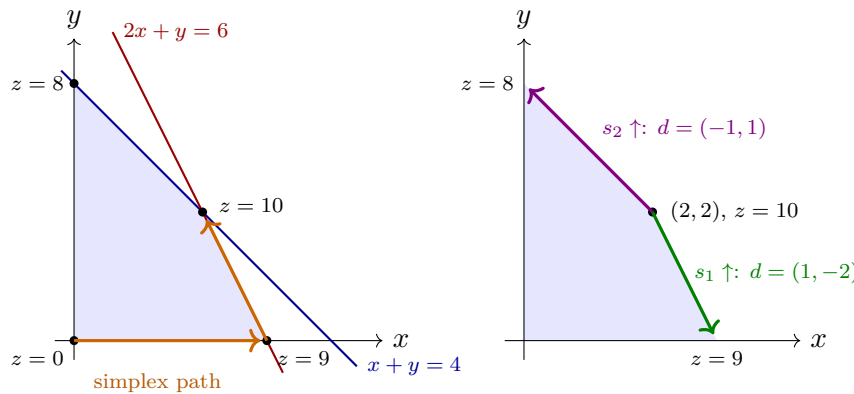
Example 5.1.1 (The running example). Two divisible activities $x, y \geq 0$ earn 3 and 2 per unit and share two resources:

$$\max 3x + 2y \quad \text{s.t.} \quad x + y \leq 4, \quad 2x + y \leq 6, \quad x, y \geq 0.$$

With slacks $s_1, s_2 \geq 0$, simplex runs as in Chapter 3. The slides call the objective z , where Chapter 3 wrote v .

Tableau 0 (basis s_1, s_2)	Tableau 1 (basis x, s_1)	Tableau 2 (basis x, y)
$s_1 = 4 - x - y$	$x = 3 - \frac{1}{2}y - \frac{1}{2}s_2$	$x = 2 + s_1 - s_2$
$s_2 = 6 - 2x - y$	$s_1 = 1 - \frac{1}{2}y + \frac{1}{2}s_2$	$y = 2 - 2s_1 + s_2$
$z = 0 + 3x + 2y$	$z = 9 + \frac{1}{2}y - \frac{3}{2}s_2$	$z = 10 - s_1 - s_2$
x in; ratios 4, 3; s_2 out	y in; ratios 6, 2; s_1 out	optimal: $z \leq 10$

The path is $(0, 0) \rightarrow (3, 0) \rightarrow (2, 2)$, with values $0 \rightarrow 9 \rightarrow 10$. In the last tableau every feasible plan has $s_1, s_2 \geq 0$, so $z = 10 - s_1 - s_2 \leq 10$: the plan $(2, 2)$ is optimal.



The right-hand picture shows the two *basic directions* at the optimum, both downhill. Section 5.2 explains where they come from.

5.2 Basic Directions: What One Simplex Move Really Is

The setting

Throughout, the LP is in equational form with m independent equations and n variables:

$$\max c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0, \quad P = \{x \in \mathbb{R}^n : Ax = b, \quad x \geq 0\}.$$

Definition 5.2.1 (Basis, BFS, nondegeneracy). Choose m column indices $B(1), \dots, B(m)$ whose columns form an invertible matrix $B = [A_{B(1)} \cdots A_{B(m)}]$, and let N be the other (nonbasic) indices. The *basic solution* sets $x_i = 0$ for $i \in N$ and $x_B = B^{-1}b$. If $x_B \geq 0$ it is a *basic feasible solution* (BFS). It is *nondegenerate* if every component of x_B is strictly positive.

Definition 5.2.2 (Feasible direction). A vector d is a *feasible direction* at $x \in P$ if $x + \theta d \in P$ for some $\theta > 0$.

One move: raise one nonbasic variable

At a BFS, pick a nonbasic $j \in N$ and let x_j rise from 0 to $\theta > 0$, keeping the other nonbasic variables at zero. So $d_j = 1$ and $d_i = 0$ for $i \in N \setminus \{j\}$. The basic variables must adjust, $x_B(\theta) = x_B + \theta d_B$, and the equations decide by how much.

You choose only one number, how much of x_j to use. The equations decide everything else.

Proposition 5.2.3 (The j th basic direction). *Staying on $Ax = b$ forces $Ad = 0$, and hence*

$$d_B = -B^{-1}A_j.$$

This $d = d^{(j)}$ is called the j th basic direction.

Proof. $A(x + \theta d) = b$ and $Ax = b$ with $\theta > 0$ give $Ad = 0$. All nonbasic entries of d vanish except $d_j = 1$, so $0 = Ad = \sum_{k=1}^m A_{B(k)}d_{B(k)} + A_j = Bd_B + A_j$. Multiply by B^{-1} . $\square$

Proposition 5.2.4 (How far we can move: the ratio test again). $x + \theta d$ stays feasible as long as $x_B + \theta d_B \geq 0$. If some $d_{B(k)} < 0$, the largest step is

$$\theta_{\max} = \min_{k: d_{B(k)} < 0} \frac{x_{B(k)}}{-d_{B(k)}}.$$

If $d_B \geq 0$, every $\theta \geq 0$ is allowed.

This is exactly Theorem 3.3.9 of Chapter 3. In tableau terms, the coefficient of x_j in the row of $x_{B(k)}$ is $d_{B(k)}$.

Reading B^{-1} off the final tableau

Proposition 5.2.5 (The tableau displays $-B^{-1}A_N$). In the equation-style tableau of a basis B , the basic rows read $x_B = B^{-1}b - B^{-1}A_N x_N$. So the column of a nonbasic x_j in the basic rows is $d_B^{(j)} = -B^{-1}A_j$. If the original slack columns form I and the slacks are all nonbasic, their columns in the tableau form $-B^{-1}$.

Proof. $Ax = b$ is $Bx_B + A_N x_N = b$. Multiply by B^{-1} and isolate x_B . For the slacks, A_N restricted to them is I . $\square$

Example 5.2.6 (B^{-1} in the running example). In Tableau 2 the slack coefficients are, row by row, $(1, -1)$ for x and $(-2, 1)$ for y . Hence

$$-B^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} \implies B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix},$$

for $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$, the columns of x and y . Check: $B^{-1}b = (-4 + 6, 8 - 6) = (2, 2)$, the optimal plan.

Gotcha

- **The sign.** Because the basic variables are isolated on the *left*, the tableau shows $-B^{-1}$, not B^{-1} . Forgetting this minus sign is the most common error in this chapter.
- **Notation clash.** Chapter 3 called the tableau constants p_i . In this chapter, following the slides, p denotes the *multiplier vector* $p^\top = c_B^\top B^{-1}$ (Section 5.3). The constants are simply “the constant column”.

5.3 Reduced Costs and the Optimality Theorem

The gain along a basic direction

Along $d^{(j)}$ the objective changes at the rate

$$c^\top d^{(j)} = c_j + \underbrace{c_B^\top d_B^{(j)}}_{\text{direct gain}} = \underbrace{c_j}_{\text{direct gain}} - \underbrace{c_B^\top B^{-1} A_j}_{\text{cost of adjusting the basis}}.$$

Using one unit of x_j earns c_j directly, but forces the basic activities to shift, and that shift costs $c_B^\top B^{-1} A_j$.

Definition 5.3.1 (Reduced cost, maximisation). The *reduced cost* of variable j for the basis B is

$$r_j = c_j - c_B^\top B^{-1} A_j = c^\top d^{(j)}.$$

In vector form, $r^\top = c^\top - c_B^\top B^{-1} A$, and $r_j = 0$ for every basic j . The *multipliers* of the basis are $p^\top := c_B^\top B^{-1} \in \mathbb{R}^m$, so that $r_j = c_j - p^\top A_j$. The reduced cost r_j is exactly the coefficient of x_j in the tableau's objective row $z = z_0 + \sum_{j \in N} r_j x_j$.

Example 5.3.2 (Two basic directions at $(2, 2)$). Order the variables (x, y, s_1, s_2) , so $c = (3, 2, 0, 0)$. From Tableau 2, the basic directions are $d^{(s_1)} = (1, -2, 1, 0)$ and $d^{(s_2)} = (-1, 1, 0, 1)$.

j	$d_B^{(j)}$	$\theta_{\max}$	reaches	$r_j = c^\top d^{(j)}$	value there
s_1	$(1, -2)$	$2/2 = 1$ (y hits 0)	$(3, 0)$	$3 - 4 = -1$	$10 - 1 = 9$
s_2	$(-1, 1)$	$2/1 = 2$ (x hits 0)	$(0, 4)$	$-3 + 2 = -1$	$10 - 2 = 8$

Both moves lose 1 per unit, matching $z = 10 - s_1 - s_2$. The multipliers are $p^\top = (3, 2)B^{-1} = (1, 1)$.

Theorem 5.3.3 (Reduced costs and optimality, maximisation). Let x be a BFS of $\max\{c^\top x : Ax = b, x \geq 0\}$ with basis matrix B .

1. If $r_j \leq 0$ for all $j \in N$, then x is optimal.
2. If x is optimal and nondegenerate, then $r_j \leq 0$ for all $j \in N$.

Proof. **(1) Compare with any feasible plan u .** From $Au = b$ we get $Bu_B + A_N u_N = b$, so $u_B = B^{-1}b - B^{-1}A_N u_N$. Since $x_B = B^{-1}b$ and $x_N = 0$, $u_B - x_B = -B^{-1}A_N u_N$. Then

$$c^\top u - c^\top x = c_B^\top (u_B - x_B) + c_N^\top u_N = (c_N^\top - c_B^\top B^{-1} A_N) u_N = \sum_{j \in N} r_j u_j.$$

Each term is (nonpositive) $\times$ (nonnegative), so $c^\top u \leq c^\top x$ for every feasible u .

(2) Suppose $r_j > 0$ for some $j \in N$. Take $d = d^{(j)}$. Since $x_B > 0$ (nondegeneracy), a small enough $\theta > 0$ keeps $x_B + \theta d_B \geq 0$, so $x + \theta d$ is feasible. Its value is $c^\top x + \theta r_j > c^\top x$, which contradicts optimality. $\square$

Key insight

The identity $c^\top u - c^\top x = \sum_{j \in N} r_j u_j$ holds for *every* solution of $Au = b$. It is the objective row of the tableau written in matrix form. Part (1) is just reading its signs.

Example 5.3.4 (Why part (2) needs nondegeneracy). Take $\max x_1$ subject to $x_1 + s = 0$, with $x_1, s \geq 0$. The only feasible point is $(0, 0)$, so it is optimal. The basis $\{s\}$ gives the degenerate BFS $s = 0$ and the tableau $s = 0 - x_1$, $z = 0 + x_1$, so $r_{x_1} = +1 > 0$. Yet no improvement is possible, because $\theta_{\max} = 0/1 = 0$. The basis $\{x_1\}$ describes the same point with $z = 0 - s$, which is a valid certificate. At a degenerate optimum, *some* basis certifies optimality, but not necessarily the one you are holding.

Definition 5.3.5 (Optimal basis matrix). B is an *optimal basis matrix* if

$$\underbrace{B^{-1}b \geq 0}_{\text{feasibility}} \quad \text{and} \quad \underbrace{r^\top = c^\top - c_B^\top B^{-1}A \leq 0^\top}_{\text{optimality}}.$$

Together these certify that $x_B = B^{-1}b$, $x_N = 0$ is optimal.

The multipliers are the dual solution

Proposition 5.3.6 (p is the optimal dual solution and the vector of shadow prices). Let $\max\{c^\top x : Ax \leq b, x \geq 0\}$ be written with slacks as $[A \ I](x, s) = b$, and let B be an optimal basis matrix. Then $p^\top = c_B^\top B^{-1}$ is an optimal solution of the dual $\min\{b^\top y : A^\top y \geq c, y \geq 0\}$ of Chapter 4, with $b^\top p =$ the optimal value.

Proof. The reduced costs of the original columns are $c - A^\top p \leq 0$, that is $A^\top p \geq c$. Those of the slack columns are $0 - I^\top p = -p \leq 0$, that is $p \geq 0$. So p is dual-feasible. Its value is $b^\top p = c_B^\top B^{-1}b = c_B^\top x_B$, the primal value. Theorem 4.2.3 makes both optimal. $\square$

This is Proposition 4.3.5 again: there, $y^* = -r_s$, and indeed $r_s = -p$. In the running example $p = (1, 1)$, and the dual value is $4 + 6 = 10$.

Summary: Simplex in matrix form

- Basic direction: $d_j = 1$, $d_B = -B^{-1}A_j$. Step: $\theta_{\max} = \min_{d_{B(k)} < 0} x_{B(k)} / (-d_{B(k)})$.
- Reduced cost: $r_j = c_j - p^\top A_j$ with $p^\top = c_B^\top B^{-1}$. It is the objective-row coefficient and the gain per unit along $d^{(j)}$.
- $r_N \leq 0$ certifies optimality; the converse needs nondegeneracy.
- The tableau's slack columns show $-B^{-1}$; p is the dual optimum.

5.4 Sensitivity Analysis: The Framework

Motivation

A factory has its optimal plan. Then the questions arrive. Is one more hour of assembly worth buying, and at what price? A product's profit changes: is the plan still best? A new product is proposed: should we make it? Re-solving from scratch would work, but the final tableau already contains B^{-1} , p and the reduced costs. *Sensitivity analysis* reuses them to answer each question locally, around the known optimum.

Optimising once is rarely the end. The data move, and the question is whether yesterday's answer still holds.

Technique: The universal recipe

Keep the optimal basis B fixed and recompute its two tests with the modified data:

$$B^{-1}b \geq 0 \quad (\text{feasibility}), \quad r^\top = c^\top - c_B^\top B^{-1}A \leq 0^\top \quad (\text{optimality}).$$

If both hold, B is still optimal and the new solution is read off immediately. If one fails, the old basis loses its certificate, and we continue pivoting from

the updated tableau.

Change	What moves in the tableau	Test that can fail	If it fails
New variable (a, γ)	a new column	$r_u = \gamma - p^\top a \leq 0$	primal simplex pivot
$b \rightarrow b + \Delta b$	constants and z_0	$B^{-1}(b + \Delta b) \geq 0$	dual simplex pivot
$c \rightarrow c + \Delta c$	objective row only	$r' \leq 0$	primal simplex pivot

5.5 Case 1: A New Activity

A proposed activity $u \geq 0$ uses the resources in column $a \in \mathbb{R}^m$ and earns γ per unit. The LP becomes $\max\{c^\top x + \gamma u : Ax + au = b, x, u \geq 0\}$. The old plan $(x^*, 0)$ stays feasible, and B is still a basis, since a new column adds no equation. Only one new reduced cost appears:

$$r_u = \gamma - c_B^\top B^{-1}a = \gamma - p^\top a.$$

Interpretation. $p^\top a$ is the value, at the current shadow prices, of the resources one unit of u would use. The new activity is worth considering only if it earns more than the resources it diverts: $\gamma > p^\top a$.

Price the new product at the shadow prices of the resources it would consume. Make it only if it earns more than that.

Example 5.5.1 (A new activity using one unit of each resource). In the running example, $a = (1, 1)$ and $p = (1, 1)$, so $p^\top a = 2$. If $\gamma = 1.5$, then $r_u = -0.5$ and the old plan stays optimal. If $\gamma = 2.5$, then $r_u = +0.5$, and the basis loses its certificate.

Continue the old tableau. Append the column $-B^{-1}a = -(0, 1) = (0, -1)$ to the basic rows and $r_u = +\frac{1}{2}$ to the objective row:

$$x = 2 + s_1 - s_2 + 0u, \quad y = 2 - 2s_1 + s_2 - u, \quad z = 10 - s_1 - s_2 + \frac{1}{2}u.$$

u enters, and $y = 2 - u$ limits it to $u \leq 2$, so y leaves. Solving the y row for u and substituting:

$$x = 2 + s_1 - s_2, \quad u = 2 - 2s_1 + s_2 - y, \quad z = 11 - 2s_1 - \frac{1}{2}s_2 - \frac{1}{2}y.$$

All objective coefficients are ≤ 0 , so the new optimum is $(x, y, u) = (2, 0, 2)$ with $z = 11$. The new activity replaces y completely.

5.6 Case 2: A Resource Level Changes

Let $b' = b + \Delta b$, with A and c fixed. The reduced costs do not involve b , so the optimality test still holds. Only feasibility can fail.

Proposition 5.6.1 (Feasibility range and shadow prices). *The basis B remains optimal if and only if $B^{-1}(b + \Delta b) \geq 0$. In that case the new optimum is $x'_B = B^{-1}(b + \Delta b)$, $x'_N = 0$, and*

$$v(b + \Delta b) = c_B^\top B^{-1}(b + \Delta b) = v(b) + p^\top \Delta b.$$

So p_i is the shadow price of resource i : the change in optimal value per extra

unit of b_i , valid while B stays feasible.

Technique: Ranging a right-hand side

For $\Delta b = \delta e_i$, the new constants are $x_B + \delta (B^{-1})_{\cdot i}$, where $(B^{-1})_{\cdot i}$ is the i th column of B^{-1} . Each row gives one linear inequality in δ ; intersect them. In the tableau, $(B^{-1})_{\cdot i}$ is *minus* the column of slack s_i .

Example 5.6.2 (Resource 1 becomes $4 + \delta$). $B^{-1}(4 + \delta, 6) = (2 - \delta, 2 + 2\delta)$. Only the constant column and $z_0 = 10 + p_1\delta = 10 + \delta$ change:

$$x = (2 - \delta) + s_1 - s_2, \quad y = (2 + 2\delta) - 2s_1 + s_2, \quad z = (10 + \delta) - s_1 - s_2.$$

Feasibility requires $2 - \delta \geq 0$ and $2 + 2\delta \geq 0$, so $\boxed{-1 \leq \delta \leq 2}$. On that range, $v(\delta) = 10 + \delta$.

Beyond the range: the dual simplex step

Outside the range, some basic variable is negative, while every objective coefficient is still ≤ 0 . The slides fix this with one pivot. The rule behind it is the *dual simplex* step:

The objective row is still a valid certificate; only feasibility broke. So pivot to repair feasibility while keeping the objective row nonpositive.

Technique: Dual simplex pivot (maximisation, equation-style tableau)

1. **Leaving variable:** a basic x_ℓ whose constant is negative, with row $x_\ell = c_\ell + \sum_{j \in N} q_{\ell j} x_j$.
2. **Entering variable:** among nonbasic j with $q_{\ell j} > 0$ (raising x_j raises x_ℓ), choose the one minimising $(-r_j)/q_{\ell j}$. This keeps every objective coefficient ≤ 0 after the pivot.
3. If no $q_{\ell j} > 0$ exists, the row says $x_\ell \leq c_\ell < 0$ for all feasible plans, so **the LP is infeasible**.

Example 5.6.3 (The whole value curve). $\delta > 2$: $x = 2 - \delta < 0$. In its row only s_1 has a positive coefficient, so s_1 enters and x leaves:

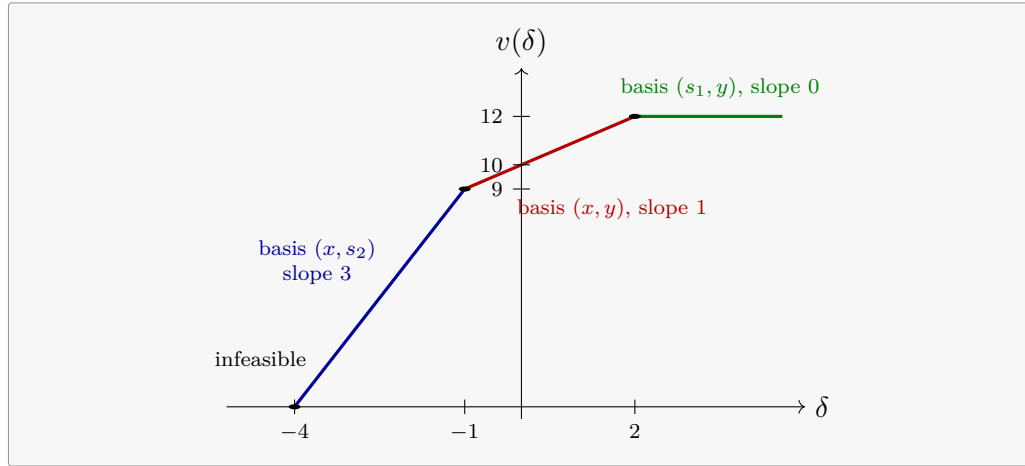
$$s_1 = (\delta - 2) + x + s_2, \quad y = 6 - 2x - s_2, \quad z = 12 - x - 2s_2.$$

This is feasible for $\delta \geq 2$, with $v = 12$: extra units of resource 1 are now worthless, since its slack is basic.

$\delta < -1$: $y = 2 + 2\delta < 0$. Only s_2 has a positive coefficient in its row, so s_2 enters and y leaves:

$$s_2 = (-2 - 2\delta) + y + 2s_1, \quad x = (4 + \delta) - y - s_1, \quad z = (12 + 3\delta) - y - 3s_1.$$

This is feasible for $-4 \leq \delta \leq -1$, with $v = 3(4 + \delta)$. For $\delta < -4$, the row $x = (4 + \delta) - y - s_1$ has no positive coefficient, so the LP is infeasible.



Interpretation & discussion

The optimal value of a maximisation LP is **piecewise linear and concave** in b . Each piece belongs to one optimal basis, and its slope is that basis's shadow price. Here the slopes are $3 \rightarrow 1 \rightarrow 0$: each extra unit of resource 1 is worth less as it becomes more plentiful. A shadow price is a *local* statement, valid only on its range.

5.7 Case 3: A Profit Changes

Let $c' = c + \Delta c$, with A and b fixed. The plan $B^{-1}b$ stays feasible, so only optimality can fail:

$$(r')^\top = (c + \Delta c)^\top - (c_B + \Delta c_B)^\top B^{-1}A \leq 0^\top?$$

- **Nonbasic coefficient** $c_j \rightarrow c_j + t$: only r_j changes, to $r_j + t$. The basis survives while $t \leq -r_j$.
- **Basic coefficient** $c_{B(k)} \rightarrow c_{B(k)} + \varepsilon$: the multipliers p change, so *every* nonbasic reduced cost may change.

Technique: Basic profit change: add ε times the row

If x_k is basic with tableau row $x_k = \bar{x}_k + \sum_{j \in N} q_{kj}x_j$, the new objective is $z + \varepsilon x_k$. Substituting the row gives the new objective row directly:

$$z' = (z_0 + \varepsilon \bar{x}_k) + \sum_{j \in N} (r_j + \varepsilon q_{kj}) x_j.$$

So the basis survives while $r_j + \varepsilon q_{kj} \leq 0$ for every nonbasic j .

Example 5.7.1 (The profit of x becomes $3 + \varepsilon$). Add $\varepsilon \times (x = 2 + s_1 - s_2)$ to $z = 10 - s_1 - s_2$:

$$z' = (10 + 2\varepsilon) + (\varepsilon - 1)s_1 - (1 + \varepsilon)s_2.$$

This agrees with $p^\top = (3 + \varepsilon, 2)B^{-1} = (1 - \varepsilon, 1 + \varepsilon)$. Optimality needs $\varepsilon - 1 \leq 0$ and $-1 - \varepsilon \leq 0$, so $\boxed{-1 \leq \varepsilon \leq 1}$. On that range the plan $(2, 2)$ stays optimal, with value $10 + 2\varepsilon$.

At $\varepsilon = 2$: $z' = 14 + s_1 - 3s_2$. Now s_1 enters, and $y = 2 - 2s_1$ limits it to

$s_1 \leq 1$, so y leaves:

$$x = 3 - \frac{1}{2}y - \frac{1}{2}s_2, \quad s_1 = 1 - \frac{1}{2}y + \frac{1}{2}s_2, \quad z' = 15 - \frac{1}{2}y - \frac{5}{2}s_2.$$

The new optimum is $(3, 0)$ with value 15.

Interpretation & discussion

Geometry. At $(2, 2)$ the binding constraints have normals $(1, 1)$ and $(2, 1)$. The vertex is optimal exactly when the objective vector $c' = (3 + \varepsilon, 2)$ lies in the cone they span. Writing $c' = \alpha(1, 1) + \beta(2, 1)$ gives $\alpha = 1 - \varepsilon$ and $\beta = 1 + \varepsilon$, which are exactly the multipliers p . Both are nonnegative if and only if $-1 \leq \varepsilon \leq 1$. At $\varepsilon = \pm 1$ the iso-profit line lies along an edge, and two vertices are optimal.

5.8 A Complete Example: Production Planning

Example 5.8.1 (The workshop's three computers). Models A, B and C use assembly time and memory boards:

	A	B	C	Available
Assembly time	1	1	1	8
Memory boards	2	1	1	10
Profit per unit	4	3	2	

$$\begin{aligned} \max \quad & 4x_A + 3x_B + 2x_C \\ \text{s.t.} \quad & x_A + x_B + x_C + s_1 = 8, \quad 2x_A + x_B + x_C + s_2 = 10, \quad x, s \geq 0. \end{aligned}$$

Solve once. The optimal basis is (x_A, x_B) , with

$$B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}, \quad B^{-1}b = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \geq 0,$$

and $p^\top = (4, 3)B^{-1} = (2, 1)$. The reduced costs are $r_C = 2 - (2 + 1) = -1$, $r_{s_1} = -2$ and $r_{s_2} = -1$, all ≤ 0 . So the optimum is $(x_A, x_B, x_C) = (2, 6, 0)$ with profit 26. Both resources are used up: $2 + 6 = 8$ and $4 + 6 = 10$. The final tableau, to be reused, is

$$x_A = 2 + 0x_C + s_1 - s_2, \quad x_B = 6 - x_C - 2s_1 + s_2, \quad z = 26 - x_C - 2s_1 - s_2.$$

The shadow prices are $p = (2, 1)$: one more assembly unit is worth 2, and one more board is worth 1, locally. Each scenario below changes the *original* model independently.

Scenario	Test	Verdict	New optimum
New model D: $a_D = (1, 1)$, $\gamma_D = 3.5$	$r_D = 3.5 - (2, 1) \cdot (1, 1) = +0.5$	basis fails; x_D enters, x_B leaves ($x_D \leq 6$)	$(x_A, x_D) = (2, 6)$, $z = 29$
Assembly $8 + \delta$	$B^{-1}(8 + \delta, 10) = (2 - \delta, 6 + 2\delta) \geq 0$	basis kept for $-3 \leq \delta \leq 2$	$z = 26 + 2\delta$; at $\delta = 1$: $(1, 8, 0)$, 28
Profit of A: $4 + \varepsilon$	row of x_A added: $r = (-1, \varepsilon - 2, -1 - \varepsilon)$	basis kept for $-1 \leq \varepsilon \leq 2$	$(2, 6, 0)$, $z = 26 + 2\varepsilon$

For model D, the pivot gives $z = 29 - \frac{1}{2}x_B - \frac{3}{2}x_C - 3s_1 - \frac{1}{2}s_2$, so the certificate is immediate. All three scenarios, and the extra ones below, were checked by solving the modified LPs exactly.

5.9 Techniques, Classic Problems and Gotchas

Technique: Exam-speed sensitivity

1. From the final tableau, read B^{-1} as *minus* the slack columns, p as *minus* the slack reduced costs, and x_B as the constants.
2. **New activity:** $r_u = \gamma - p^\top a$. Its tableau column is $-B^{-1}a$.
3. **RHS change** δe_i : new constants $x_B + \delta B^{-1}e_i \geq 0$ give the range; the value is $v + p_i\delta$ inside it.
4. **Nonbasic profit change** t : keep the basis while $t \leq -r_j$.
5. **Basic profit change** ε : add $\varepsilon \times$ (its row) to the objective row; keep the basis while all coefficients are ≤ 0 .
6. **Outside a range:** an infeasible constant calls for a dual simplex pivot; a positive objective coefficient calls for an ordinary pivot.

Classic problems

- *Production example: boards become $10 + \delta$. Range and value?* $B^{-1}(8, 10 + \delta) = (2 + \delta, 6 - \delta) \geq 0$, so $-2 \leq \delta \leq 6$, with $z = 26 + \delta$ (slope $p_2 = 1$). At $\delta = 6$: $(8, 0, 0)$, 32.
- *Assembly becomes 11 ($\delta = 3$, beyond the range)?* $x_A = -1 + s_1 - s_2$ is infeasible, and only s_1 has a positive coefficient, so s_1 enters and x_A leaves: $x_B = 10 - 2x_A - x_C - s_2$, $z = 30 - 2x_A - x_C - 3s_2$. The optimum is $(0, 10, 0)$ with 30, and extra assembly time is now worth 0.
- *A product E with $a = (1, 2)$ earning 3?* $r_E = 3 - (2 \cdot 1 + 1 \cdot 2) = -1$: do not make it. It breaks even at a profit of 4.
- *How profitable must model C become before it is worth making?* $r_C = -1$, so its profit must exceed $2 + 1 = 3$.
- *Running example: shadow price and range of resource 2?* $p_2 = 1$, and $B^{-1}(4, 6 + \delta) = (2 + \delta, 2 - \delta)$, so $-2 \leq \delta \leq 2$.

Gotcha

- The tableau shows $-B^{-1}$ in the slack columns, and $-p$ as the slack reduced costs. Two minus signs, both easy to lose.
- These are **maximisation** conventions: $r_j > 0$ means improving. Many textbooks (for example Bertsimas–Tsitsiklis) minimise, and there $\bar{c}_j < 0$ means improving. Translate before comparing.
- A shadow price holds only on its range. Beyond it, the basis changes and so does the slope.
- A change in a *basic* profit moves p , so every reduced cost must be rechecked, not just one.
- Changing b never breaks optimality of the reduced costs; changing c never breaks feasibility. Know which test each change can affect.
- Part (2) of Theorem 5.3.3 fails at degenerate points (Example 5.3.4).

Notes on the original slides

Every tableau, range and pivot on the Lecture 5 slides was recomputed, both by hand and by solving each modified LP exactly. All are correct. Points worth knowing:

- **Slides 10 and 20** make the same point (B^{-1} from the slack columns) twice; Proposition 5.2.5 states it once, in general.
- **Slides 34–35** repair infeasibility by choosing the leaving variable first, the negative basic one. This is the *dual simplex* method, which the slides use without naming. The entering rule is in the technique box of Section 5.6.
- **Slide 15**, part 2, needs nondegeneracy, and Example 5.3.4 shows why. The slides prove it but give no counterexample.
- **Slide 24** introduces $p^\top = c_B^\top B^{-1}$ as “resource multipliers”. These are exactly the optimal dual variables of Chapter 4 (Proposition 5.3.6).
- **Slide 36**: the plot’s vertical axis marks 9 and 12. The curve also passes through $(0, 10)$ and reaches 0 at $\delta = -4$.

Chapter takeaways**Summary: Simplex tableau and sensitivity**

- A simplex move follows the basic direction $d_B = -B^{-1}A_j$; its gain per unit is the reduced cost $r_j = c_j - p^\top A_j$.
- B is optimal when $B^{-1}b \geq 0$ and $r \leq 0$; then $p^\top = c_B^\top B^{-1}$ is the optimal dual solution, the shadow prices.
- New activity: $r_u = \gamma - p^\top a$. New b : keep B while $B^{-1}(b + \Delta b) \geq 0$; value $v + p^\top \Delta b$. New c : keep B while $r' \leq 0$.
- Outside a range, pivot from the saved tableau (dual simplex for a negative constant, primal for a positive r_j). The value is piecewise linear and concave in b .